

Lab - Blob Snapshots and Versioning

To Begin with the Lab

Summary

This lab demonstrates how to protect blob data using **Blob Snapshots** and **Blob Versioning** in Azure Storage. Snapshots create read-only copies of a blob at a specific point in time, while Versioning automatically keeps previous versions whenever a blob is modified or deleted, making it easy to recover data.

Understanding

Azure Blob Storage provides two important data protection features:

- **Blob Snapshots** create a **read-only copy** of a blob at a specific point in time. Snapshots are created manually and remain available until they are deleted. They are useful before making changes to important files because they allow you to restore the original content if needed.
- **Blob Versioning** automatically creates a **new version** every time a blob is modified or overwritten. Each version has a unique version ID, allowing you to view, restore, or download previous versions without manually creating snapshots.

Using snapshots and versioning together helps prevent accidental data loss, enables recovery from unwanted changes, and supports backup and auditing requirements.

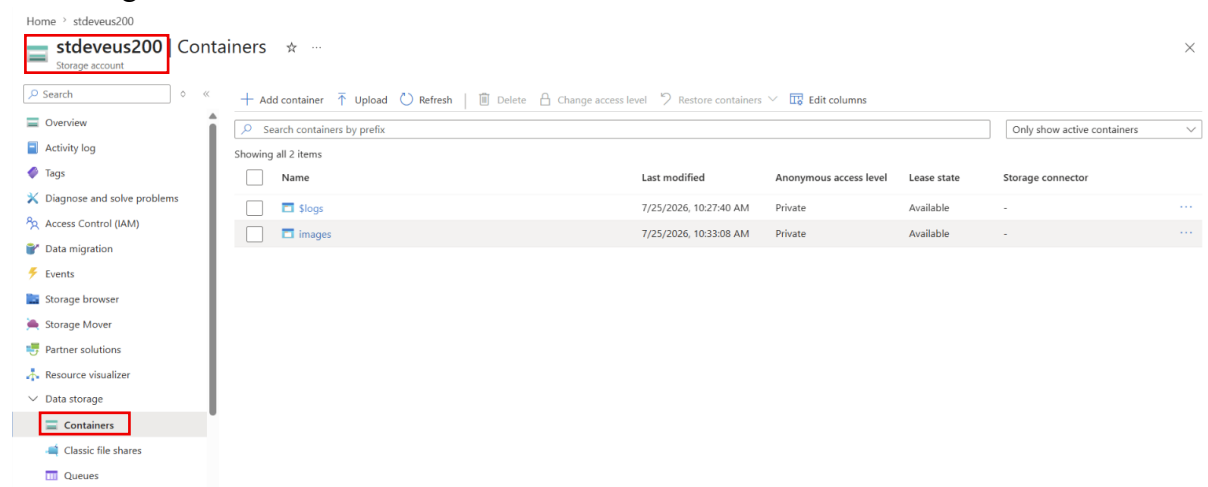
Example:

Suppose a document named **report.pdf** is stored in Azure Blob Storage. Before editing it, you create a snapshot. Later, another user updates the file. Since Blob Versioning is enabled, Azure automatically saves the previous version. If the latest version contains incorrect information, you can restore either the snapshot or a previous version.

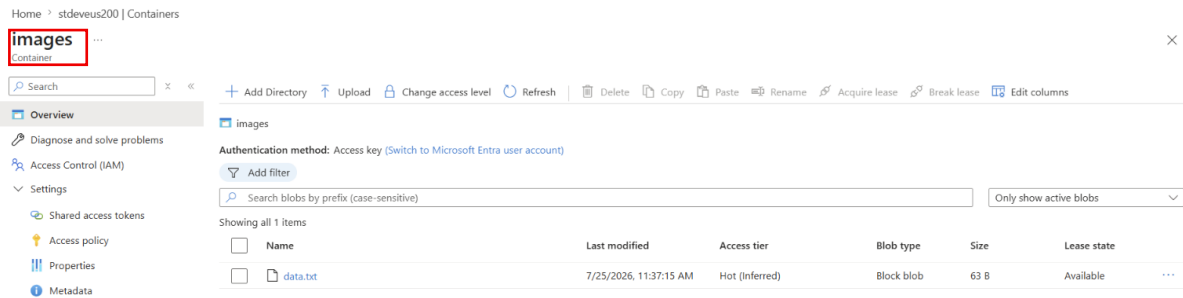
Lab Steps

Step 1: Open the Storage Account

- Sign in to the Azure Portal.
- Open your **Storage Account**.
- Navigate to **Containers**.

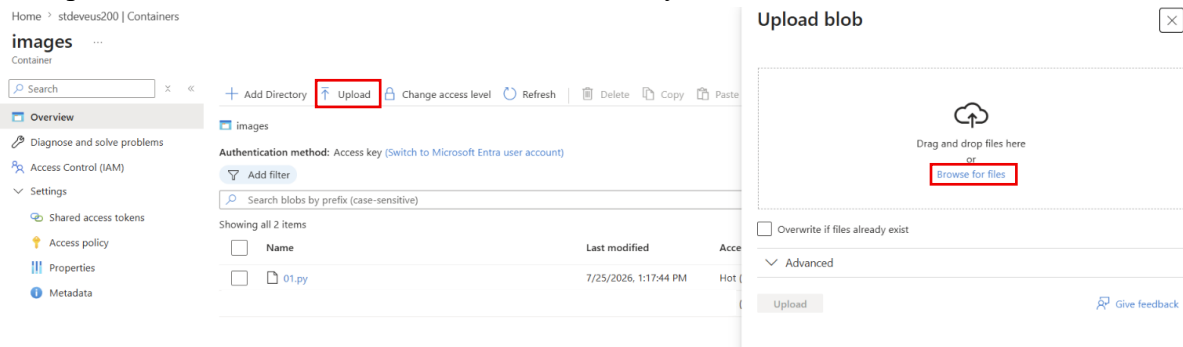


- Open the **container(images)** containing the blob.

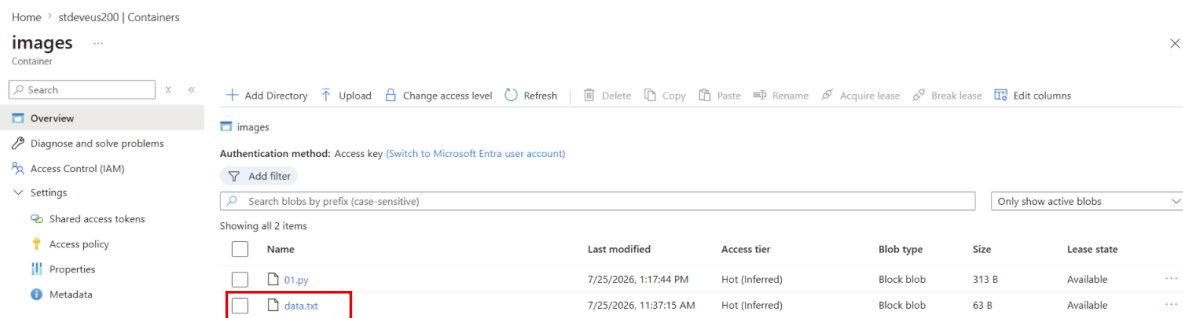


Step 2: Upload a Blob

- Click on the **Upload**
- Upload a file to the container if one is not already available.

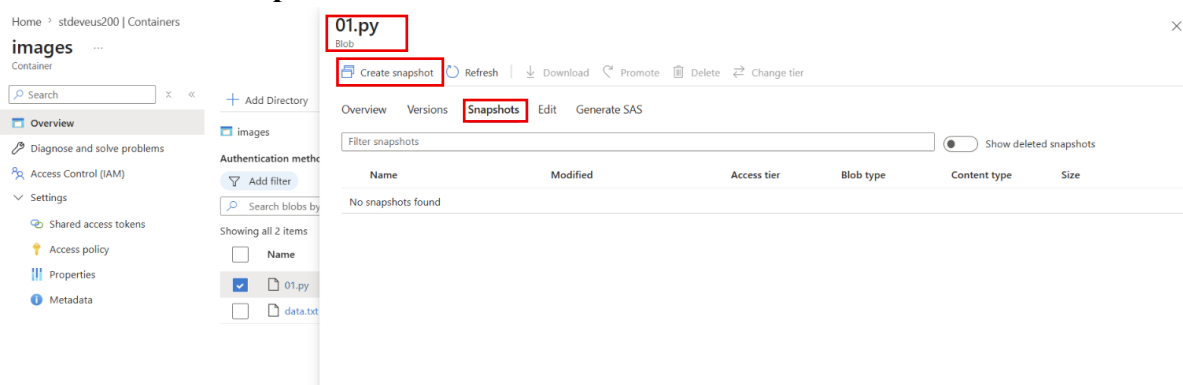


- Verify that the blob appears in the container.



Step 3: Create a Blob Snapshot

- Select the blob.
- Navigate to **Snapshot**.
- Click **Create Snapshot**.



- Azure creates a read-only snapshot of the blob.
- Verify that the snapshot appears under the blob.

01.py
×

Create snapshot
Refresh
Download
Promote
Delete
Change tier

Overview
Versions
Snapshots
Edit
Generate SAS

Filter snapshots
Show deleted snapshots

Name	Modified	Access tier	Blob type	Content type	Size	
☐ 01.py (7/25/2026, 1:28:47 PM)	7/25/2026, 1:17:44 PM	Hot	Block blob	application/octet-str...	313 B	...

Step 4: Modify the Blob

- Upload a new version of the same file.

01.py
×

Save
Discard
Download
Refresh
Delete

Overview
Versions
Snapshots
Edit
Generate SAS

```

1 fruit = []
2
3 f1 = input("Enter your fruit name0111: ")
4 fruit.append(f1)
5 f2 = input("Enter your fruit name: ")
6 fruit.append(f2)
7 f3 = input("Enter your fruit name: ")
8 fruit.append(f3)
9 f4 = input("Enter your fruit name: ")
10 fruit.append(f4)
11 f5 = input("Enter your fruit name: ")
12 fruit.append(f5)
13
14 print(fruit)

```

- Choose **Overwrite** when prompted.

01.py
×

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Name	Modified	Access tier	Blob type	Content type	Size	
☒ 01.py (7/25/2026, 1:38:45 PM)	7/25/2026, 1:38:15 PM	Hot	Block blob	application/oc		...

Download
Promote
Properties
Delete
Change tier

- Verify that the blob content has changed.

```
01.py
Blob

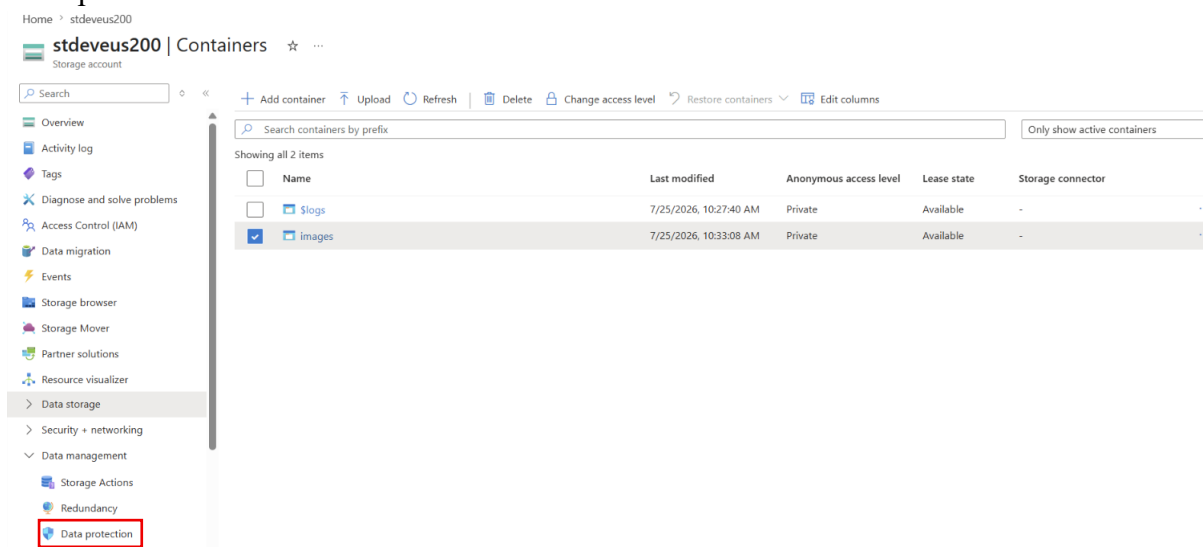
Save Discard Download Refresh Delete

Overview Versions Snapshots Edit Generate SAS

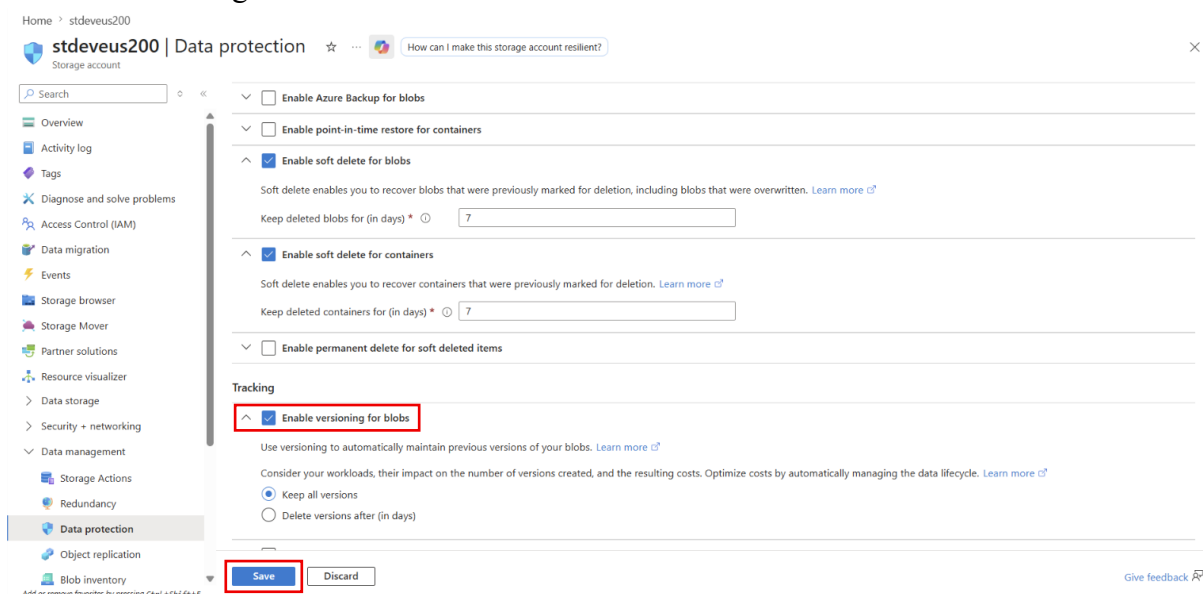
1 fruit = []
2
3 f1 = input("Enter your fruit name0111: ")
4 fruit.append(f1)
5 f2 = input("Enter your fruit name: ")
6 fruit.append(f2)
7 f3 = input("Enter your fruit name: ")
8 fruit.append(f3)
9 f4 = input("Enter your fruit name: ")
10 fruit.append(f4)
11 f5 = input("Enter your fruit name: ")
12 fruit.append(f5)
13
14 print(fruit)
```

Step 5: Enable Blob Versioning

- Return to the **Storage Account**.
- Open **Data Protection**.



- Enable **Blob Versioning**.
- Save the changes.



Step 6: Upload a New Version

- Return to the container.
- Upload the same file again with updated content.
- Overwrite the existing blob.
- Azure automatically creates a new blob version.

01.py

Blob

 Save  Discard  Download  Refresh  Delete

Overview Versions Snapshots Edit Generate SAS

```
1 fruit = []
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```

Step 7: View Blob Versions

- Select the blob.
- View the list of available versions.
- Observe that each version has a unique Version ID and creation time.

01.py

Blob

 Refresh  Download  Make current version  Delete  Change tier

Overview Versions Snapshots Edit Generate SAS

When blob versioning is enabled, you can restore an earlier version of a blob to recover your data. Every write operation to a blob in the account results in the creation of a new version. Consider your workloads, their impact on the number of versions created, and the resulting costs. Optimize costs by automatically managing the data lifecycle. [Learn more](#)

Filter versions

☒ Show deleted versions

Version Id	Modified	Access tier	Blob type	Size	
☐ 2026-07-25T08:09:09.1963750Z	7/25/2026, 1:39:09 PM	Hot	Block blob	317 B	***

Step 8: Restore a Previous Version

- Select an older version of the blob.
- Restore or promote it as the current version.

01.py

Blob

Refresh Download Make current version Delete Change tier

Overview Versions Snapshots Edit Generate SAS

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Filter versions

Show deleted versions

Version Id	Modified	Access tier	Blob type	Size
☒ 2026-07-25T08:09:09.1963750Z	7/25/2026, 1:39:09 PM	Hot	Block blob	

Download

Make current version

Generate version SAS

Delete

Change tier

Access policy

- Verify that the blob content has been successfully recovered.

01.py

Blob

Refresh Download Make current version Delete Change tier

Overview Versions Snapshots Edit Generate SAS

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Filter versions

Show deleted versions

Version Id	Modified	Access tier	Blob type	Size	
☐ 2026-07-25T08:17:43.7795443Z	7/25/2026, 1:47:43 PM	Hot	Block blob	321 B	...
☐ 2026-07-25T08:09:09.1963750Z	7/25/2026, 1:39:09 PM	Hot	Block blob	317 B	...

Step 9: Access Blob Snapshots

- View the snapshots associated with the blob.
- Compare the snapshot with the current blob.

01.py

Blob

Save Discard Download Refresh Delete

Overview Versions Snapshots Edit Generate SAS

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10 fruit.append(f4)
11 f5 = input("Enter your fruit name: ")
12 fruit.append(f5)
13
14 print(fruit)
```

Step 10: Verify Data Recovery

- Confirm that both snapshots and versions can be used to recover previous data.

- Ensure the correct version is available after restoration.

ui.py
Blob

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 Refresh |  Download  Make current version  Delete  Change tier

Overview Versions Snapshots Edit Generate SAS

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Filter versions ☒ Show deleted versions

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